A Comparative Study of Pedestrian Footfall Trends using Time Series forecasting

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Abstract. The study of urban population dynamics has long been an important area of research. Forecasting pedestrian footfall is crucial for urban management, public health, crime analysis, and evaluating the influence of public gatherings. This research delves into the dynamics of urban population to accurately anticipate pedestrian numbers in specific locations and times, leveraging historical data for comprehensive analysis and predictions. This research focuses on predicting pedestrian footfall in Dublin city using various machine learning models tailored for time series analysis. Examining three distinct models—Holt-Winters, SARIMA (Seasonal Autoregressive Integrated Moving Average), and Prophet—we aim to forecast pedestrian numbers. Our study evaluates the efficacy of these models in accurately predicting footfall, contributing insights into their suitability for forecasting pedestrian traffic. The models were evaluated using RMSE (Root Mean Square Error) and the results shows that for the given dataset SARIMA performed well. An explanation of these methods, along with limitations, criteria for choosing and further considerations and suggestions of work are discussed.

Keywords: Pedestrian footfall, Time series, Forecasting, Trend, Seasonality

1. Introduction

The analysis of population dynamics has long been a cornerstone of diverse fields, spanning urban planning, transportation, public health, and safety. A significant volume of pedestrians signifies the crowdedness of the environment, which is also influenced by multiple urban variables such as weather, lighting [1], the dependability of public transportation services[2]. Instances of heightened human density necessitate prompt attention for resource planning and allocation to meet escalating pedestrian needs. Hence there is a tremendous need for both internal and external stakeholders to continue adopting a long-term strategy for urban planning and management. This includes enhancing city walkways and transportation, evaluating event influences on pedestrian activity, devising marketing strategies for optimal exposure, and identifying staffing and resource needs. These mentioned reasons could be broadly categorized as[3] Urban Management, i.e. understanding footfall aids in evaluating the impact of development initiatives, guiding decisions to enhance city centres and optimize urban spaces; Crime Analysis, i.e Quantifying the influence of residents and visitors on crime rates helps in comprehending safety concerns and devising strategies for crime prevention ; Event Success Measurement , i.e Accurate footfall prediction determines the success of public events in attracting crowds to cities, influencing future event planning and impact assessment; Development analysis , ie Comparative analysis of footfall changes over time illuminates whether a city centre is thriving or declining, offering early warnings for targeted investment and preventing worsening situations.

There are few studies conducted with different pedestrian footfall dataset from different cities, where different Machine Learning Techniques are used. Our focus here is to first understand the data and then perform modeling using appropriate models matching to the data.The case study area is the city of Dublin, Ireland, where abundant pedestrian data exist thanks to a large number of sensors that have been installed by the local government and reported on a publicly available open data portal [10].

The structure of this paper is outlined as follows: Section 2 presents a comprehensive review of pertinent

literature, establishing the research objectives and anticipated results. Methodology specifics are detailed in Sections 3, with risk and ethical considerations in Section 4. Section 5 offers the experiments and evaluation of machine learning models, followed by detailed discussions and conclusions in Section 6. Finally, Section 7 articulates the research's contributions to the field.

So our focus of this paper is to understand if machine learning models can be used to analyse and predict pedestrian footfall data, and also to further understand various ways to choose the right model.As our dataset is a time series one, we are using appropriate time series models for data analysis and forecasting. The study consider three different time series forecasting models and compare their evaluation results to decide on the best model for our data set. The time series models considered in this project are Holt-Winters, SARIMA (Seasonal Auto regressive Integrated Moving Average), and Facebook(meta)' s Prophet. The various phases of the study include, writing a proposal, performing background research, searching for appropriate data, analysing the data, building and evaluating the model, and creating the report.

2. Literature Review

In the previous section we have seen why it is important to understand the pedestrian footfall in a city. In this paper we try to formulate a system that can help in forecasting the pedestrian footfall data. With our understanding and of the problem and the data, we believe machine learning techniques could be a good choice for our problem. In this literature review we will first have a general understanding of the data we have then as per that we formulate our research question on the type of machine learning techniques to use .

2.1. Dataset

The Dublin City Council pedestrian footfall dataset [10] has Pedestrian footfall counts of people at a number of locations in Dublin city. Passersby are counted and logged every hour, 24 hours per day, 7 days per week using a network of PYRO-Box(Sensor) people counters located throughout central Dublin. As the dataset is capturing only hourly data and the pedestrain count during that time, we can consider our use case as an example pf time series data. A time series is a collection of data points gathered over a period of time and ordered chronologically [4].

2.2. Related works

Though we dont have any similar work on pedestrian footfall analysis for Dublin city, there are few other researches done in City of Melbourne [5]. The authors used a dataset from Melbourn which had more details like Station name, latitude and longitude of each sensor. The study mainly focused on just one time series model-ARIMA. There is one more study[6] for the same city of Melbourne with a different dataset. The dataset used is more cross sectional in nature with details like Street furniture, Buildings, Landmarks around each sensor. Due to the cross sectional nature of the dataset the study focused more on modeling the relationships between the variables using non linear models like XGBoost and Random Forest. There are few other footfall analysis researches which were done in context of retail stores[7], recreational walking trails[8] to understand the peoples behaviour. But the dataset used in most of the earlier works were different from our case where we do not have any other features to form a relationship.Since our dataset has only timestamp and corresponding pedestrian count, we focus on machine learning techniques to handle such datasets.

2.3. Time series machine Learning models

Time series analysis is a specific way of analyzing a sequence of data points collected over time. In Time series analysis, analysts record data points at consistent intervals over a set period rather than just recording the data points intermittently or randomly.There are different components for a time series analysis- Trend, Seasonality, Cyclical, Irregularity. The details of this are covered in section 3.4. It is also important to note that we might need time series specific techniques in the dataprocessing step also. For example, for time series data there are unique methods for missing data handling and train/test data splitting. They are covered in more detail in the Methodology section.

Our initial examination of our dataset shows that it has got a sequence of data points collected over time. So we formulate our first research question around that; *How to choose between various time series models based on the data set we have ?*. Section 3.4 covers how we look at our dataset to decide on various models that we believe to solve our problem.

Once we finalize on the relevant time series techniques, the next research question is *How various time series models can be used to predict the trends in Pedestrian Footfall dataset?*. We fit our dataset into

different models we have choosen and then analyse the results and compare them. Then we come to a conclusion if any(or all) models can be used to model our dataset to predict the pedestrain footfall. To help us evaluate the performance of each forecasting model, we need to measure the differences between predicted values and the actual or the observed values. For the models, we used the commonly used measurement metric root mean square error(RMSE).

3. Methodology

Due to the nature of time-series data, there are a number of specificities involved in time series modeling that are not relevant to other datasets. So it is important to understand the data we have before deciding on any of the modeling methods, especially with time series data set. In the following sections we will have a closer look at the Data set we have and different data processing steps we have taken in our project.

3.1. Data Extraction and Description

The data set used for this research paper is obtained from a publicly available repository of Dublin City Council. The repository has different data sets and the one we used is Pedestrian Footfall DCC[10], which captures the Pedestrian footfall across Dublin city.It has the record of pedestrian footfall counts of people at a number of locations in Dublin city for years from 2016. Passersby are counted and logged every hour, 24 hours per day, 7 days per week using a network of PYRO-Box people counters located throughout central Dublin.

We have taken the footfall data for the years 2021, 2022 and 2023(till November 1st). Each data set has a timestamp field column in DD-MM-YYY HH:MM format , and 102 other columns for different streets in Dublin city. Each of those columns capture the hourly footfall count corresponding to the time stamp. On further examining of the data set we could see 102 streets in the data set have many redundant information, i.e.there are three columns corresponding to a single street - one with street name , one with street name 'IN' and another with street name 'OUT'. While examining the values of those three columns, we could find a relationship between them as footfall for *street* = footfall for *street IN* + footfall for *street OUT*, as shown in the Figure 1. We have handled such redundancy, and that will be covered in 3.2 .

3.2. Data Processing

This step explains the various processing we done on the data to make them for modeling. As mentioned in 3.1, we have 3 different data set files in .csv format corresponding to three years. We initially examined each of those files individually and did some processing before combining them to a final file.

The first important step carried out on individual data set was to remove the earlier mentioned redundancy (3.1). For every street the corresponding columns *street IN* and footfall for *street OUT* were dropped, and after that our data set columns got reduced by 1/3rd from 102 streets to 34 streets. This step was carried out on three individual data sets.

The next important step was to compare and see all the three datasets have the same street names. This is important as we would eventually need to merge the three years data into a single dataset. On our examination we could see the street names were same across all three dataset, but had some minor differences in their naming. For example, 2021 dataset had a street name *Dawson Street/Molesworth* where as the corresponding street name in 2023 had minor difference as *Dawson Street/Molesworth Pedestrian* . We corrected such differences, so that all the three dataset now have the same street names.

Then we merged the three datasets to get a single dataset with combined footfall data from years 2021, 2022, 2023(till November 1). Now our merged dataset have 24840 timestamps (hourly time stamps from 2021 January 1st 12:00 AM to 2023 November 1st 11:00 PM), and corresponding pedestrian footfall counts for 34 streets.

3.3. Combined Data Processing

Once we get the combined dataset, it is important to examine the it for missing data. Unfortunately, our dataset had a lot of streets with missing values. For example for street *O'Connell St/Pennys Pedestrian* for 24840 timestamps only 8334 had some footfall counts. There were many more streets with such huge number of missing values. As our project aims at understanding the pedestrian footfall trend across Dublin city as a whole, the first approach we have taken is to remove the streets drop the streets with more number of missing values. This is because, applying some missing values imputation techniques to those many streets might bring some bias into the data, and we might then miss the properties of the original dataset.

Date and Time	Aston Quay/Fitzgeralds	Aston Quay/Fitzgeralds IN	Aston Quay/Fitzgeralds OUT	Bachelors walk/Bachelors way	Bachelors walk/Bachelors way IN	Bachelors walk/Bachelors way OUT
01-01-2021 00:00:00	165.0	65.0	100.0	66.0	19.0	47.0

Fig. 1. Example from Pedestrian Footfall dataset.

In order to drop such streets, we set a threshold value of 24000, which is almost 3% of the total timestamp count. We then dropped all streets with entries less than the value of that threshold. Eventually we got 13 streets which had at least 24000 entries in them (Figure 2).

Though we selected streets with maximum data, still many of them have missing values(Figure 2). So handling such missing values is the next step in our process. The approach we used for handling those missing values is a similar concept of two technique used for missing value imputation in time series dataset called - Last observation carried forward (LOCF) and Next observation carried backward (NOCB) . These methods replace missing values either with the immediately preceding observed value (LOCF) or the subsequent observed value (NOCB). As few streets in our dataset had so many continuous missing values just following NOCB or LOCF might not fill the whole missing data(for example for *street O'Connell st/Princes st North* there are continuous consecutive missing values from indices 10157 to 10162). Our approach to fill the missing data is on assumption that a missing footfall/pedestrian count for an hour can have relation to ;previous hour (or next hour) footfall count (LOCF/NOCB) and same hour data of same day in the previous week (or next week), i.e, 168 hours before and after .

Using this approach we imputed missing values in our dataset. For example Figure 3 and Figure 4 shows the before and after data plot for street *O'Connell st/Princes st North*.

Next step in our dataset processing is to resampling the dataset. Resampling involves changing the frequency of your time series observations and there are two types of resampling in time series data; Upsampling: increasing the frequency of the samples, such as from minutes to seconds and Downsampling: decreasing the frequency of the samples, such as from days to months. As our dataset is in hourly timestamp and we are intrested in the daily footfall analysis, we use downsampling technique to group hourly data to daily data. After downsampling we have our finalprocessed dataset with 1035 days and corresponding daily footfall count for 13 streets(Figure 5). As we are intrested in the footfall trend in the whole Dublin city , the final step in our data pre processing is to combine all streets to get the total daily pedestrian footfall(Figure 5) .

3.4. ML Modeling

Before we do any model selection for a time series problem it is very important to understand the dataset we have. At this stage it is also handy to understand few important terms in time series analysis. A time series dataset is said to have mainly three components -trend, seasonality, and residual. Understanding these components helps in understanding underlying patterns in the data, forecasting, and making business or economic decisions. Trend can be understood as the underlying direction in which the data is moving, either upward, downward, or stable over time. It represents the long-term progression of the series. Seasonality is the Seasonal patterns that repeat at regular intervals within a certain period (e.g., monthly, quarterly, yearly). This component captures the effects of seasonality, such as increased sales during the holiday season. Residual is the remainder of the time series after accounting for the trend and seasonality. It includes random fluctuations and noise, as well as any patterns that are not captured by the trend and seasonal components.

In order to decide on the models for our problem, we first examine the data mainly Trend and Seasonality. We will use Time Series Decomposition technique to extract multiple types of variation from your dataset-

#	Column	Non-Null Count
0	Date and Time	24840 non-null
1	Baggot st upper/Mespil rd/Bank	24479 non-null
2	College Green/Bank Of Ireland	24707 non-null
3	College st/Westmoreland st	24687 non-null
4	D'olier st/Burgh Quay	24662 non-null
5	Grafton Street / Nassau Street / Suffolk Street	24400 non-null
6	Grafton Street/CompuB	24632 non-null
7	Grand Canal st upp/Clanwilliam place	24352 non-null
8	North Wall Quay/Samuel Beckett bridge East	24836 non-null
9	North Wall Quay/Samuel Beckett bridge West	24825 non-null
10	O'Connell st/Princes st North	24822 non-null
11	Phibsborough Rd/Enniskerry Road	24726 non-null
12	Richmond st south/Portabello Harbour inbound	24826 non-null
13	Richmond st south/Portabello Harbour outbound	24799 non-null

Fig. 2. The Footfall dataset after filtering streets with more missing values.

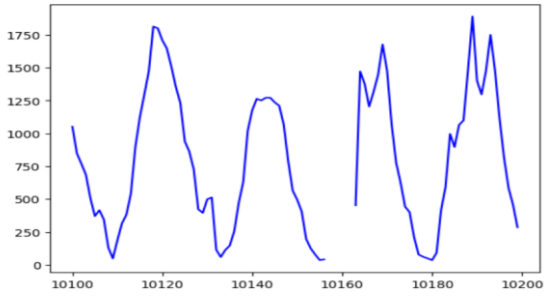


Fig. 3. Street *O'Connell st/Princes st North*- before Missing value Imputation.

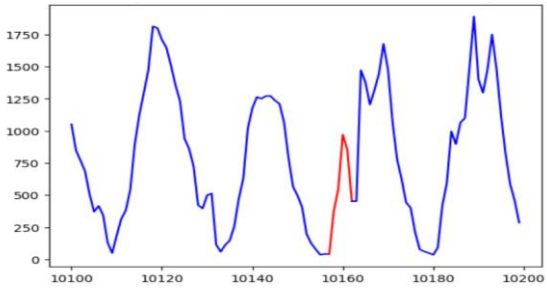


Fig. 4. Street *O'Connell st/Princes st North*- after Missing value Imputation.

seasonality, trend, and noise, and the result is as in Figure 6.

Here we can see overall increasing trend which becomes almost steady by the end of 2022 , with an yearly seasonality .There are lot of models available for time series dataset. But choosing the appropriate model depends on the nature of the dataset we are dealing with. Mainly we need to consider if our dataset have components like stationary , Seasonality , Trend etc.. Our footfall dataset is not stationary , and has got seasonality and Trend .

So for our further analysis we will use and compare three models which is suitable for datatset with Seasonality and Trend - Holt- Winters, SARIMA, and Prophet.

Holt-Winters is a model suitable for time series data with trend and/or seasonal components. This model takes into account the same seasonality trends, and instead utilizes exponential smoothing (more recent historical data important in forecasting than older data). For this model we also need to identify the *frequency of seasonality* or m. Our daily footfall dataset can have daily , weekly or yearly seasonality , but we can ignore daily seasonality for now and consider weekly and yearly seasonality. So we model the two cases separately by changing the parameter *frequency of seasonality* (weekly: m= 7, yearly:m= 365).

Seasonal AutoRegressive Integrated Moving Average (SARIMA) belongs to the ARIMA (AutoRegressive Integrated Moving Average)family which captures moving averages throughout time series, taking into account the data's trend, seasonality, and noise. In other words, ARIMA models look at differences between values in the time series. SARIMA builds upon the concept of ARIMA but extends it to model the seasonal elements in your data. SARIMA includes several parameters that can be tuned to achieve optimal performance. They are- Trend Elements:p (Trend autoregression order), d (Trend difference order), q (Trend moving average order) and Seasonal Elements: P(Seasonal autoregressive order), D(Seasonal difference order), Q(Seasonal moving average order), m(The number of time steps for a single seasonal period). In order to get the best prediction, it's important to find the values of SARIMA(p,d,q)(P,D,Q)m that optimize a metric of interest. Here we use grid search to iteratively explore different combinations of parameters. The evaluation metric we'll use for the grid search is the AIC (Akaike Information Criterion) value. The AIC measures how

Column	Non-Null Count
Baggot st upper/Mespil rd/Bank	1035 non-null
College Green/Bank Of Ireland	1035 non-null
College st/Westmoreland st	1035 non-null
D'olier st/Burgh Quay	1035 non-null
Grafton Street / Nassau Street / Suffolk Street	1035 non-null
Grafton Street/CompuB	1035 non-null
Grand Canal st upp/Clanwilliam place	1035 non-null
North Wall Quay/Samuel Beckett bridge East	1035 non-null
North Wall Quay/Samuel Beckett bridge West	1035 non-null
O'Connell st/Princes st North	1035 non-null
Phibsborough Rd/Enniskerry Road	1035 non-null
Richmond st south/Portabello Harbour inbound	1035 non-null
Richmond st south/Portabello Harbour outbound	1035 non-null
Total_Footfall	1035 non-null

Fig. 5. Final Daily Pedestrian footfall data.

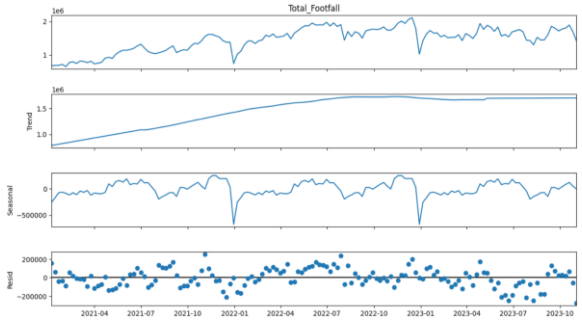


Fig. 6. Time Series Decomposition -to extract Trend , Seasonality , Noise.

well a model fits the data while taking into account the overall complexity of the model. In general, we want to pick the combination with the lowest AIC value. Also for SARIMA we use only the weekly frequency of seasonality, as per Rob J Hyndman[9] suggestion to use frequency attribute to 7 When we have a daily dataset with lot of records(few Years).

Prophet is Facebook(meta)'s open source time series model.The simplicity of this model makes for plug-and-play of Data Analysts and Scientists alike. A user simply has to pass a dataframe, with two columns - column 'ds' which contains the datetime values and 'y' the number of observations. A big difference between Prophet and the other two models discussed is that Prophet allows for multi-seasonality forecasting with daily, weekly and yearly along with holiday detection.

To help us evaluate the performance of each forecasting model, we need to measure the differences between predicted values and the actual or observed values. For the mentioned models, we used the commonly used measurement metric root mean square error(RMSE) also referred to as root-mean-square deviation (RMSD).

4. Risks and Ethical Consideration

In this project there are risks and ethical considerations we need to take, and the first and foremost is to understand that similar to any machine learning technique, the model we propose might not be perfect and it could produce wrong results and analysis .

One risk we would particularly want to understand is about the risk in our Data quality.As mentioned in the methodology section, many streets in our dataset have missing values, and so we had to drop many of them. This could bring limitations to our aim of understanding the pedestrian footfall trend in whole Dublin city. Also there could be the risk of related back end hardware. For few streets in in our data set, we could see for many days/weeks the footfall count was very low and all of a sudden they increased and them maintained a steady high count. These type of scenarios could be related to the hardware risk related to the back end sensors.

Along with the risk, we should take in account of some of the ethical consideration when doing a similar research on Pedestrian Footfall.

One important consideration should be around the fairness of the system. This could be related to the earlier mentioned data quality issue. When we remove many streets from our dataset, the fairness of the whole dataset might be lost. So we need to take appropriate measures to ensure the data quality and fairness is preserved .

Another ethical consideration to keep in mind is about the transparency. The research process and the technology developed should be transparent, with clear explanations of the data sources, algorithms, and decision-making processes. The researchers needs to keep in mind that they should be accountable for the accuracy and reliability of the results.

5. Experiments and evaluation

5.1. Data Splitting

Before we fit the datset into the mentioned models, we need to divide the data into training and test datasets. This is mainly valid for the first two algorithms Holt-Winters and SARIMA. From our dataset of 1035 days, we will first 847 days for training and remaining 188 days for testing.

5.2. Holt-Winters Modeling

For Holt-Winters we fit and experiment the above dataset with two seasonalities, weekly and yearly. The result of the fitting of the two cases are shown in Figure 7 and Figure 8. From the figures we can see both the seasonalities do not give a good prediction. The RMSE value obtained for weekly and yearly seasonality for Holt-Winters are 45095 and 79155 respectively .

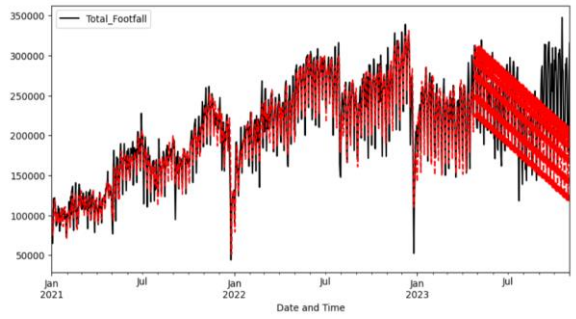


Fig. 7. Holt Winters Forecasting Visualization- Weekly Seasonal Frequency

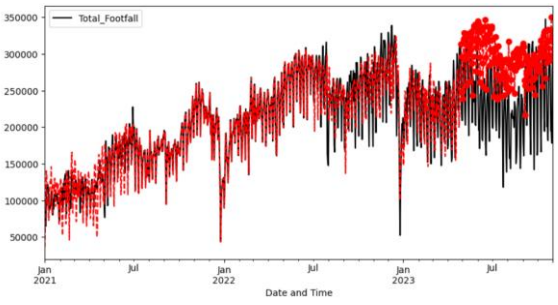


Fig. 8. Holt Winters Forecasting Visualization- Yearly Seasonal Frequency

5.3. SARIMA Modeling

As mentioned in section 3.4 we model SARIMA using weekly seasonal frequency only. Also as mentioned we first performed a grid search to iteratively explore different combinations of parameters-(p,d,q)(P,D,Q)m. The grid search tested all possible combinations of variables, and gave the set that resulted in the lowest AIC. In our experiment we got SARIMA(1,0,1)x(0,1,1,7) has the lowest AIC value(23159.036). Then we use this parameters and the dataset to fit our model.

The result of the fitting gave a better result compared to the previous Holt-Winters method. Figure 9 shows the prediction plot and we can see SARIMA model best captures both the seasonality and trend of our dataset. The RMSE value obtained for SARIMA is 24163 which also better compared to Holt Winters.

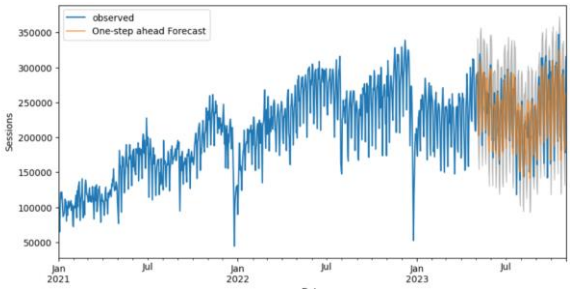


Fig. 9. SARIMA Forecasting Visualization- Weekly Seasonal Frequency

5.4. Prophet Modeling

One difference of prophet from other two models is that we do not need to mention the seasonality frequency, as the model itself takes care of it. But as mentioned in section 3.4 we would need to do rename the dataframe columns to 'ds' and 'y' before fitting it to the model. For our dataset Prophet gave a good fit with RMSE value 38137. Figure 10 shows the plot of the prediction and we can see Prophet also captures the seasonality and trend of our dataset.

The Table 1 shows the combined results of our three models, and we can see SARIMA gave a better result for our dataset compared to Holt-Winters and Prophet.

Table 1
Evaluation Score

Model	Season Length	RMSE
Holt-Winters'	7 days	45095
	365 days	79155
SARIMA	7 days	24163
Prophet	-	38137

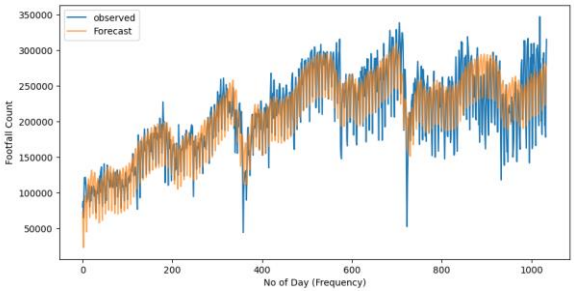


Fig. 10. Prophet Forecasting Visualization

6. Conclusion and Future Work

In conclusion the study has shown that time series forecasting models can be used to model and forecast Pedestrian Footfall. In our case of Dublin City data,the SARIMA model gives better result compared to Prophet and Holt Winters with RMSE value of 24163 . SARIMA could capture the Trend and Seasonality well in the dataset.But considering the extended capabilities of Prophet like its ability to accommodate multiple seasonality , and even its ability to account additional Regressors and holiday or unusual patters, Prophet is also a good candidate for forecasting like daily footfall analysis . The study also revealed that choosing a right model for time series dataset depends on the components of the dataset.

For future work this project can be extended to integrate the effect of weather, or Holidays on Footfall Data and other models which can accomodate multiple seasonalities and regressors at the same time can be considered. There also Prophet can be considered as a good candidate.

Furthermore, a streetwise analysis of the the Pedestrian footfall analysis can be considered to unveil the patterns within different areas of the city and it could be used for better City planning

7. Contributions

Overall contributions of this research to the field can be summed up as follows:

Three Time Series models were applied to the dataset and it is concluded that SARIMA performed well in our dataset.

Data processing could be considered as an important step in Time series modeling, and understanding the dataset is a cruicial step in deciding on the Models to choose. For any time series dataset the appropriate model has to be chosen based on the components of datset mainly Trend and Seasonality.

The Pedestrian Footfall forecasting could be used in many areas like Urban Management, Crime Analysis, Event Success Measurement etc.. and eventually for a better Urban life.

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